

Perturbation Theory for Stochastic Differential Equations 9

Abstract

In this chapter we want to combine the perturbative method developed in Chap. 4 with the functional representation of stochastic differential equations introduced in Chap. 7. The Ornstein–Uhlenbeck process studied as a special case in Chap. 8 in this context plays the role of the solvable, Gaussian part of the theory. We here want to show how to calculate perturbative corrections that arise from non-linearities in the stochastic differential equation, corresponding to the non-Gaussian part of the action.

9.1 Vanishing Moments of Response Fields

We now would like to extend the system from the previous section to the existence of a non-linearity in the stochastic differential equation (8.1) of the form

$$dx(t) = f(x(t)) dt - \tilde{j}(t) dt + dW(t), \quad (9.1)$$

where $f(x)$ is some non-linear function of x . We first want to show that, given the value of the source $j = 0$, all moments of the response field vanish. In the derivation of the path-integral representation of Z in Chap. 7, we saw that Z belongs to a properly normalized density, as demonstrated by (7.5), so $Z[j = 0] = 1$. The same normalization of course holds in the presence of an arbitrary value of $\tilde{j}$ in Eq. (9.1), because $\tilde{j}$ corresponds to an additional term on the right-hand side of the stochastic differential equation and our derivation of Z holds for any right-hand side. As a consequence we must have

$$Z[0, \tilde{j}] \equiv 1 \quad \forall \tilde{j}. \quad (9.2)$$

We hence conclude that any derivative by $\tilde{j}$ of (9.2) must vanish, so that all moments of $\tilde{x}$ vanish

$$\frac{\delta^n}{\delta \tilde{j}(t_1) \cdots \delta \tilde{j}(t_n)} Z[0, \tilde{j}] = \langle \tilde{x}(t_1) \cdots \tilde{x}_n(t_n) \rangle \equiv 0 \quad \forall n > 0.$$

We note that the latter condition holds irrespective of the value of $\tilde{j}$; we may also evaluate the moments of $\tilde{x}$ at some non-zero $\tilde{j}$, corresponding to a particular value of the inhomogeneity on the right-hand side of Eq. (9.1).

We also note that this argument does not rely on any perturbation expansion; it is an exact property that follows directly from the structure of the action, namely the equivalence of the source field $\tilde{j}$ to an additive term in the deterministic function f . In the literature, one also finds perturbative arguments why moments of the response field vanish. For completeness, this argument is reviewed in Sect. 9.5.

9.2 Feynman Rules for SDEs in Time Domain and Frequency Domain

An arbitrary given action first needs to be converted into algebraic form in the fields, typically by Taylor expansion. We then have a stochastic differential equation with a linear, solvable part and an additional non-linearity, for example

$$dx(t) + x(t) dt - \frac{\alpha}{2!} x^2(t) dt = dW(t). \quad (9.3)$$

The action is therefore $S[x, \tilde{x}] = S_0[x, \tilde{x}] - \frac{\alpha}{2!} \tilde{x}^T x^2$, where $S_0[x, \tilde{x}] = \tilde{x}^T (\partial_t + 1)x + \frac{D}{2} \tilde{x}^T \tilde{x}$ is the Gaussian part. After having determined the propagators corresponding to the Gaussian part S_0 of the action, given in frequency domain by (8.11) with $m = -1$,

$$\begin{aligned} \Delta(t, s) &= \begin{pmatrix} \langle x(t)x(s) \rangle & \langle x(t)\tilde{x}(s) \rangle \\ \langle \tilde{x}(t)x(s) \rangle & \langle \tilde{x}(t)\tilde{x}(s) \rangle \end{pmatrix} = \begin{pmatrix} \overleftrightarrow{x(t) \quad x(s)} & \overleftarrow{x(t) \quad \tilde{x}(s)} \\ \overrightarrow{\tilde{x}(t) \quad x(s)} & 0 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2D} e^{-|t-s|} & -H(t-s) e^{-(t-s)} \\ -H(s-t) e^{-(s-t)} & 0 \end{pmatrix}, \end{aligned} \quad (9.4)$$

For problems that are time-translation invariant, often a formulation in Fourier domain leads to simpler expressions. By help of Sect. 8.3, we transfer the Feynman rules from time to frequency domain. We first express the interaction vertex in terms of the Fourier transforms of the fields to get

$$\begin{aligned}
 \xrightarrow{\mathcal{F}} \quad \tilde{X}(\omega_1) \quad \text{---} \quad \begin{array}{c} X(\omega_2) \\ \diagup \\ \text{---} \\ \diagdown \\ X(\omega_3) \end{array} &= -\frac{\alpha}{2!} \iiint \frac{d\omega_1}{2\pi} \frac{d\omega_2}{2\pi} \frac{d\omega_3}{2\pi} \\
 &\quad \times \underbrace{\int dt e^{i(\omega_1+\omega_2+\omega_3)t}}_{2\pi \delta(\omega_1+\omega_2+\omega_3)} \tilde{X}(\omega_1) X(\omega_2) X(\omega_3) \\
 &= -\frac{\alpha}{2!} \iint \frac{d\omega_1}{2\pi} \frac{d\omega_2}{2\pi} \tilde{X}(\omega_1) X(\omega_2) X(-\omega_1 - \omega_2).
 \end{aligned}$$

So we get from the Dirac- δ that the frequencies at each vertex need to sum up to zero. We may therefore think of the frequency “flowing” through the vertex and obeying a conservation equation—the frequencies flowing into a vertex also must flow out. We note that we get one factor $(2\pi)^{-1}$ less than the number of legs of the vertex. The number of factors $(2\pi)^{-1}$ therefore equals the number of remaining momentum integrals.

Moreover, we see that every external leg comes with a factor $(2\pi)^{-1}$ from the integration over ω and a factor 2π from the connecting propagator, so that the overall number of such factors is not affected. Each propagator connecting an internal pair of X or $\tilde{X}$ comes, by (8.11), with a factor 2π . Due to the conservation of the frequencies also at each propagator by the Dirac- δ , in the final expression there are hence as many frequency integrals left as we have factors $(2\pi)^{-1}$. We may therefore also only keep a single frequency dependence of the propagator and write the term $2\pi \delta$ explicitly, hence defining

$$2\pi \delta(\omega + \omega') \Delta(\omega) := \Delta(\omega, -\omega'), \quad (9.6)$$

to get the matrix of propagators

$$\begin{aligned}
 2\pi \delta(\omega + \omega') \Delta(\omega) &= \begin{pmatrix} \langle X(\omega) X(\omega') \rangle & \langle X(\omega) \tilde{X}(\omega') \rangle \\ \langle \tilde{X}(\omega) X(\omega') \rangle & 0 \end{pmatrix} \\
 &= 2\pi \delta(\omega + \omega') \begin{pmatrix} (-i\omega - 1)^{-1} D(i\omega - 1)^{-1} (-i\omega - 1)^{-1} \\ (i\omega - 1)^{-1} & 0 \end{pmatrix}.
 \end{aligned} \quad (9.7)$$

As an example, the first-order correction to the first moment then has the form

$$\begin{aligned}
 J(-\omega) \text{ (loop)} &= \int \frac{d\omega}{2\pi} J(-\omega) \int \frac{d\omega'}{2\pi} 2\pi \delta(\omega + \omega') \Delta_{x\tilde{x}}(\omega) \\
 &\times \iint \frac{d\omega_1}{2\pi} \frac{d\omega_2}{2\pi} \frac{-\alpha}{2!} 2\pi \delta(\omega' + \omega_1 + \omega_2) 2\pi \delta(\omega_1 + \omega_2) \Delta_{xx}(\omega_1) \\
 &= \int \frac{d\omega}{2\pi} J(-\omega) 2\pi \delta(\omega) \Delta_{x\tilde{x}}(\omega) \frac{-\alpha}{2!} \iint \frac{d\omega_1}{2\pi} \Delta_{xx}(\omega_1) \\
 &= \int \frac{d\omega}{2\pi} J(-\omega) 2\pi \delta(\omega) (-i\omega - 1)^{-1} \frac{-\alpha}{2!} \\
 &\times \int \frac{d\omega_1}{2\pi} (-i\omega_1 - 1)^{-1} D(i\omega_1 - 1)^{-1} \\
 &= J(0) \frac{\alpha}{2!} \int \frac{d\omega_1}{2\pi} (-i\omega_1 - 1)^{-1} D(i\omega_1 - 1)^{-1},
 \end{aligned} \tag{9.8}$$

where the connecting external line $\leftarrow = (-i\omega - 1)^{-1}$ has the shown sign, because $J(-\omega)$ couples to $X(\omega)$, so we need to take the upper right element in (9.7). The last factor $\longleftrightarrow = \Delta_{xx}(\omega_1) = (-i\omega_1 - 1)^{-1} D(i\omega_1 - 1)^{-1}$ is the covariance function connecting the two $X(\omega_1)$ and $X(\omega_2)$ legs of the vertex.

Since originally each integral over ω_i comes with $(2\pi)^{-1}$ and each conservation of sums of ω in either a propagator or a vertex comes with $2\pi \delta(\sum_i \omega_i)$, we have as many factors $(2\pi)^{-1}$ as we have independent momentum integrals. We summarize the rules as follows:

- An external leg ending on $J(\omega)$ attaches to a variable $X(-\omega)$ within the diagram and analogous for $\tilde{J}(\omega)$ and $\tilde{X}(-\omega)$.
- At each vertex, the sum of all ω flowing into the vertex must sum up to zero, since we get a term $\propto \delta(\sum_{i=1}^n \omega_i)$.
- The frequencies that enter a propagator line must also exit, since we get a term $\propto \delta(\omega + \omega')$.
- We have as many factors $(2\pi)^{-1}$ as we have independent ω integrals left after all constraints of ω -conservation have been taken into account.
- The number of ω integrals hence must correspond to the number of loops: all other frequencies are fixed by the external legs.

So we may infer the frequencies on each propagator line by rules analogous to Kirchhoff's law: Treating the frequencies as if they were conserved currents. Using these rules we could have written down the fourth line in Eq. (9.8) directly.

The above integral by

$$\begin{aligned}
 & \frac{1}{2\pi} \int d\omega_1 (-i\omega_1 - 1)^{-1} D(i\omega_1 - 1)^{-1} \\
 &= \frac{1}{2\pi i} \int_{-i\infty}^{i\infty} dz (-z - 1)^{-1} D(z - 1)^{-1} \\
 &= \frac{1}{2\pi i} \int_{\gamma} dz (-z - 1)^{-1} D(z - 1)^{-1} \\
 &= \frac{D}{2}
 \end{aligned} \tag{9.9}$$

hence evaluates to $\frac{\alpha}{2} \frac{D}{2} J(0)$. We here closed the path γ in the positive direction, which is the left half-plane (with $\Re(z) < 0$), we get a +1 from the winding number. We encircle the pole $z = -1$ from the left factor and need to replace $z = -1$ in the right term. An additional minus sign comes from $(-z - 1)^{-1} = -(z + 1)^{-1}$.

The result, being proportional to $J(0)$, shows that the correction only affects the stationary expectation value at $\omega = 0$, which therefore is (by the functional chain rule)

$$\langle x(t) \rangle = \left. \frac{\delta W}{\delta j(t)} \right|_{j=0} = \int \underbrace{e^{-i\omega t}}_{\frac{\delta J(\omega)}{\delta j(t)}} \left. \frac{\delta \hat{W}}{\delta J(\omega)} \right|_{J=0} d\omega = \int e^{-i\omega t} \delta(-\omega) \frac{\alpha D}{4} d\omega = \frac{\alpha D}{4},$$

which is valid to first order in α and which is of course the same as (9.5), obtained in time domain. We here used that due to the source being of the form $J^T X = \int \frac{d\omega}{2\pi} J(-\omega) X(\omega)$ that $\frac{\delta \hat{W}}{\delta J(\omega)} = \frac{1}{2\pi} 2\pi \frac{\alpha D}{4} \delta(-\omega)$.

9.3 Diagrams with More Than a Single External Leg

In calculating diagrams with more than a single external leg, we remember that the n -fold repetition of an external leg must come from the factor

$$\exp(j^T x) = \sum_n \frac{(j^T x)^n}{n!}.$$

So a diagram with n -legs of identical type j comes with n time-integrals and a factor $n!^{-1}$. This is completely analogous to the case of the n -fold repetition of an interaction vertex.

It is instructive to first derive the correction to W —hence we compute the j -dependent contribution—and only in a second step differentiate the result by j to obtain the correction to the cumulants.

For example, a diagram contributing to the correction of the variance of the process would come with a factor $\frac{1}{2!} \iint dt ds j(t) j(s)$ prior to taking the second derivative by j . Concretely, let us consider the diagram

$$\begin{aligned}
 & 2 \cdot 2 \cdot 2 \cdot \begin{array}{c} \Delta_{xx} \\ \xrightarrow{j(t)} \text{---} \text{---} \text{---} \Delta_{xx} \\ \Delta_{\tilde{x}x} \end{array} \xrightarrow{j(s)} \Delta_{\tilde{x}x} \\
 &= \frac{1}{2!} \iint dt ds j(t) j(s) \\
 & \quad \times \underbrace{2 \cdot 2 \cdot 2 \cdot \frac{1}{2!} \left(\frac{\alpha}{2!}\right)^2 \int dt' ds' \Delta_{xx}(t, t') \Delta_{xx}(t', s') \Delta_{\tilde{x}x}(t', s') \Delta_{\tilde{x}x}(s', s)}_{=: f(t, s)} \\
 &=: \frac{1}{2!} \iint dt ds j(t) j(s) f(t, s).
 \end{aligned}$$

The combinatorial factor arises from two possibilities of connecting the Δ_{xx} propagator of the left external leg to either of the vertices and the two possibilities of choosing the incoming x -leg of the vertex to which we connect this external leg. Another factor two arises from the two possibilities of connecting the Δ_{xx} propagator to either of the two x -legs of the right vertex. All other contractions are uniquely determined then; so in total we have a factor $2 \cdot 2 \cdot 2$.

In calculating the contribution to the covariance function $\langle\langle x(t)x(s) \rangle\rangle$, the second cumulant of the process, we need to take the second functional derivative. Because the factor j appears twice, we obtain by the application of the functional product rule the correction to $\delta^2 W / \delta j(t) \delta j(s)$

$$\frac{1}{2!} (f(t, s) + f(s, t)), \quad (9.10)$$

which is a manifestly symmetric contribution as it has to be for a covariance function. A single term $f(t, s)$ is not necessarily symmetric, as seen from the appearance of the non-symmetric functions $\Delta_{\tilde{x}x}$.

We may calculate the same contribution in frequency domain. To assign the frequencies to the legs we use that at each line the frequencies must have opposite sign on either end and the sums of all frequencies at a vertex must sum up to zero; the frequency of the left field of the propagator is the argument of the corresponding

function $\Delta(\omega)$, according to (9.6). So we get

$$\begin{aligned}
 I &:= 2 \cdot 2 \cdot 2 \cdot \begin{array}{c} \text{Diagram: A loop with two vertices. The left vertex has an incoming arrow from the left labeled } J(\omega) \text{ and an outgoing arrow to the left labeled } -\omega \Delta_{xx} \omega. \text{ The right vertex has an incoming arrow from the right labeled } J(-\omega) \text{ and an outgoing arrow to the right labeled } -\omega \Delta_{\tilde{x}x} \omega. \text{ The top arc of the loop is labeled } (-\omega + \omega') \Delta_{xx} (\omega - \omega'). \text{ The bottom arc is labeled } -\omega' \Delta_{\tilde{x}x} \omega'. \end{array} \\
 &= \frac{1}{2!} \int \frac{d\omega}{2\pi} J(\omega) \\
 &\quad \times \underbrace{\int \frac{d\omega'}{2\pi} 2 \cdot 2 \cdot 2 \cdot \frac{1}{2!} \left(\frac{\alpha}{2!}\right)^2 \Delta_{xx}(-\omega) \Delta_{xx}(-\omega + \omega') \Delta_{\tilde{x}x}(-\omega') \Delta_{\tilde{x}x}(-\omega) J(-\omega)}_{=: F(\omega)} \\
 &=: \frac{1}{2!} \int \frac{d\omega}{2\pi} J(\omega) F(\omega) J(-\omega). \tag{9.11}
 \end{aligned}$$

We observe that the contribution can be written as an integral over one frequency, the frequency within the loop ω' . Each of the sources is attached by a propagator to this loop integral. We will see in the following that the inner integral thus behaves as an effective two-point vertex—what is known as a 1PI vertex, to be introduced in Chap. 11.

The contribution to the variance therefore becomes with the functional chain rule and $\delta J(\omega)/\delta j(t) = e^{-i\omega t}$

$$\frac{\delta^2 W}{\delta j(t) \delta j(s)} = \iint d\omega d\omega' e^{-i\omega t} e^{-i\omega' s} \frac{\delta^2 W}{\delta J(\omega) \delta J(\omega')}.$$

By the last line in (9.11) and the application of the product rule we see that

$\frac{\delta^2 I}{\delta J(\omega) \delta J(\omega')} = \frac{1}{2!} \frac{1}{2\pi} \delta(\omega + \omega') (F(\omega) + F(-\omega))$ so that

$$\frac{\delta^2 I}{\delta j(t) \delta j(s)} = \int \frac{d\omega}{2\pi} e^{i\omega(s-t)} \frac{1}{2!} (F(\omega) + F(-\omega)). \tag{9.12}$$

Again, the product rule causes a symmetric contribution of the diagram. The back transform can be calculated with the help of the residue theorem. Multiple poles of order n can be treated by Cauchy's differential formula

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint \frac{f(z)}{(z-a)^{n+1}} dz.$$

9.4 Appendix: Unitary Fourier Transform

A unitary transform is defined as an isomorphism that preserves the inner product. In our example the space is the vector space of all functions and the inner (scalar)

product is

$$(f, g) = \int_{-\infty}^{\infty} f^*(t) g(t) dt. \quad (9.13)$$

The Fourier transform is a linear mapping of a function $f(t)$ to $F(\omega)$, which can be understood as the projection onto the orthogonal basis vectors $u_\omega(t) := \frac{1}{2\pi} e^{i\omega t}$. The basis is orthogonal because

$$(u_\omega, u_{\omega'}) = \int_{-\infty}^{\infty} \frac{e^{i(\omega' - \omega)t}}{(2\pi)^2} dt = \frac{\delta(\omega' - \omega)}{2\pi}. \quad (9.14)$$

The Fourier transform is a unitary transformation, because it preserves the form of the scalar product on the two spaces

$$\begin{aligned} (f, g) &:= \int f^*(t) g(t) dt = \int d\omega \int d\omega' \underbrace{\int dt \frac{e^{i(-\omega + \omega')t}}{(2\pi)^2}}_{\equiv \delta(-\omega + \omega')/(2\pi)} F^*(\omega) G(\omega') \\ &= \int \frac{d\omega}{2\pi} F^*(\omega) G(\omega) =: (F, G). \end{aligned} \quad (9.15)$$

So the scalar products in the two spaces have the same form.

Changing the path integral from $\int \mathcal{D}x(t)$ to $\int \mathcal{D}X(\omega)$, each individual time integral can be expressed by all frequency integrals as

$$\int dx(t) = \int d\omega \frac{e^{i t \omega}}{2\pi} \int dX(\omega). \quad (9.16)$$

The transform (9.16) is a multiplication with the (infinite dimensional) matrix $U_{t\omega} = \frac{e^{i t \omega}}{2\pi}$. This matrix $U \equiv (u_{-\infty}, \dots, u_\infty)$ has the property

$$(U^{T*}U)_{\omega\omega'} \stackrel{(9.14)}{=} \frac{\delta(\omega - \omega')}{2\pi},$$

which is the infinite dimensional unit matrix, from which follows in particular that $|\det(U)| = \text{const}$. Hence changing the path integral $\int \mathcal{D}x(t)$ to $\int \mathcal{D}X(\omega)$ we only get a constant from the determinant. Since we are only interested in derivatives of generating functionals, this constant has no consequence. However, the integration boundaries change. The integral $\int \mathcal{D}x(t)$ goes over all real-valued functions $x(t)$. Hence the corresponding Fourier transforms $X(\omega)$ have the property $X(-\omega) = X(\omega)^*$.

The action in (8.3) instead of the standard scalar product on $\mathbb{C}$ (9.13) employs the Euclidean scalar product between functions x and y of the form

$$x^T y = \int dt x(t) y(t).$$

As a consequence, in frequency domain we get

$$\int \frac{d\omega}{2\pi} X(-\omega) Y(\omega). \quad (9.17)$$

9.5 Appendix: Vanishing Response Loops

We would like to treat the non-linear function $f(x)$ in (9.1) perturbatively, so we consider its Taylor expansion $f(x(t)) = f^{(1)}(0)x(t) + \sum_{n=2}^{\infty} \frac{f^{(n)}(0)}{n!} x(t)^n$. We here restrict the choice of f to functions with $f(0) = 0$, because an offset can be absorbed into a non-vanishing external source field $\tilde{j} \neq 0$. For clarity of notation we here treat the one-dimensional case, but the extension to N dimensions is straightforward. We may absorb the linear term in the propagator, setting $m := f^{(1)}(0)$ as in the linear case (8.2). The remaining terms yield interaction vertices in the action

$$S[x, \tilde{x}] = \underbrace{\tilde{x}^T (\partial_t - f^{(1)}(0)) x + \tilde{x}^T \frac{D}{2} \tilde{x}}_{S_0[x, \tilde{x}]} - \underbrace{\sum_{n=2}^{\infty} \frac{f^{(n)}(0)}{n!} \tilde{x}^T x^n}_{V[x, \tilde{x}]},$$

which are of the form

$$V[x, \tilde{x}] = \sum_{n=2}^{\infty} \frac{f^{(n)}(0)}{n!} \underbrace{\tilde{x}^T x^n}_{\int \tilde{x}(t) x^n(t) dt} = \begin{array}{c} \tilde{x}(t) \swarrow \nearrow x(t) \\ \leftarrow \quad \rightarrow \\ \dots \end{array}, \quad (9.18)$$

where the ellipsis indicates the remaining legs attached to an x , one leg for each power in x .

We saw in the previous section that the response functions in the Gaussian case are causal, i.e. $\langle x(t) \tilde{x}(s) \rangle = 0$ for $t \leq s$ and also that $\langle \tilde{x}(t) \tilde{x}(s) \rangle = 0 \quad \forall t, s$. We will now show that this property is conserved in presence of an arbitrary non-linearity that mediates a causal coupling. To this end consider a perturbative correction to the response function with a single interaction vertex. Since the interaction vertices are of the form (9.18), they couple only equal time arguments (see underbrace in Eq.(9.18)). A contribution to a response function $\langle x(t) \tilde{x}(s) \rangle$ requires a bare propagator $\rightarrow$ from $\tilde{x}(s)$ to one of the three right legs of the vertex (9.18) and one additional propagator $\rightarrow$ from the left leg of the vertex to the external $x(t)$. The

remaining x -legs of the vertex need to be contracted by the propagator $\longleftrightarrow$. Since both propagators to the external legs mediate a causal interaction and the vertex forces the intermediate time points of both propagators to be identical, it implies that the correction is unequal zero only for $t > s$. We also see from this argument that a generalization to causal interactions is straightforward.

By the inductive nature of the proof of connectedness in Sect. 5.2, this argument holds for arbitrary orders in perturbation theory, since the connected diagrams with $i + 1$ vertices are formed from those with i : If causality holds for response functions with i vertices, this property obviously transcends to order $i + 1$ by the above argument, hence it holds at arbitrary order.

The same line of arguments shows that all correlators of the form $\langle \tilde{x}(t) \cdots \tilde{x}(s) \rangle = 0$ vanish. We know this property already from the general derivation in Sect. 9.1, which only required the normalization condition and of course holds for arbitrary non-linearities f . Often one finds in the literature diagrammatic arguments for the vanishing moments, which we will show here for completeness.

Indeed, at lowest order, the form of (8.14) shows that second moments of $\tilde{x}$ vanish. The first moment of $\tilde{x}$, by differentiating (8.7) by $\delta Z / \delta \tilde{j}(t)|_{\tilde{j}=0} = \langle \tilde{x}(t) \rangle = \int \Delta_{\tilde{x}(t)x(s)} j(s) ds$ as well vanishes for $j = 0$, which even holds for $\tilde{j} \neq 0$ due to the absence of $\Delta_{\tilde{x}\tilde{x}} = 0$. The independence of $\tilde{j}$ is consistent with the possibility to absorb the source term $\tilde{j}$ in the inhomogeneity of the differential equation.

In the non-linear case, corrections to the mean value would come from graphs with one external $\tilde{j}$ leg. Such a leg must be connected by the response function $\rightarrow$ to one of the x -legs of the vertex, so that again a free $\tilde{x}(t)$ leg of the vertex remains. Due to the vanishing mean $\tilde{x}$, we only have the option to connect this free leg to one of the $x(t)$ -legs of the vertex by another response function. We still get a vanishing contribution, because response functions (in the here considered Ito convention) vanish at equal time points, $\langle x(t)\tilde{x}(t) \rangle = 0$ (see Sect. 7.5), and all time points of fields on the interaction vertex are identical. The generalization to general causal relationships, i.e. $\tilde{x}(t), x(s)$ with $t \geq s$ on the vertex, holds analogously. The same property holds in the Stratonovich convention, as outlined below.

The same argument holds for all higher moments of $\tilde{x}$, where for each external line $\tilde{j}$ one propagator $\rightarrow$ attaches to the corresponding x -legs of the vertex. The remaining single $\tilde{x}$ -leg of the vertex again cannot be connected in a way that would lead to a non-vanishing contribution. The argument generalizes to higher order corrections, by replacing the bare propagators by the full propagators, which, by the argument given above, have the same causality properties.

Comparing this result to the literature [1, see p. 4914 after eq. (9)] and [2, see eq. (7)], a difference is that these works considered the Stratonovich convention. An additional term $-\frac{1}{2}f'(x)$ is present in the action (7.19), because the Stratonovich convention amounts to $\alpha = \frac{1}{2}$. The response function at zero time lag then is $\langle x(t)\tilde{x}(t) \rangle = \frac{1}{2}$ (see Sect. 7.5). The contributions of loops closed by response

functions $\langle \tilde{x}(t)x(t) \rangle$ that end on the same vertex of the form (9.18) are

$$\begin{aligned}
 \text{Diagram} &= n \underbrace{\langle x(t)\tilde{x}(t) \rangle}_{=\frac{1}{2}} \frac{f^{(n)}}{n!} x(t)^{n-1} \\
 &= \frac{1}{2} \partial_x \frac{f^{(n)}}{n!} x(t)^n,
 \end{aligned}$$

where two of the n x -legs are shown explicitly. The wiggly line with the ellipsis indicates the $n - 1$ remaining legs of the interaction vertex. The combinatorial factor n in the first line stems from the n possible ways to attach the propagator to one of the n factors x of the vertex. The last line shows that the remaining term is the opposite of the contribution $-\frac{1}{2} f'(x)$ that comes from the functional determinant in (7.19). In conclusion, all contributions of closed loops of response functions on the same vertex are canceled in the Stratonovich convention in the same way as in the Ito convention.

9.6 Problems

(a) Extension to N Dimensions

Convince yourself that the action of an N -dimensional Ornstein–Uhlenbeck process with correlated white noise and covariance matrix D_{ij} is indeed given by Eq. (8.2). (2 points).

(b) Linear Differential Operators, Propagators, Gaussian Integrals

In this exercise we want to review the connection between Gaussian integrals, propagators (Green's functions), and the inversion of symmetric matrices. To illustrate these connections let us assume the linear first-order homogeneous differential equation

$$\frac{dx}{dt} - m x = 0.$$

We want to calculate the Green's function in various ways. First, solve the defining equation for the fundamental solution of the differential operator $\frac{d}{dt} - m$ also called the Green's function Δ

$$\frac{d\Delta(t)}{dt} - m \Delta(t) = \delta(t)$$

analytically. It is sufficient to state what the solution is. Show that a particular solution of the inhomogeneous equation

$$\frac{dx}{dt} - m x = f(t)$$

is given by

$$x(t) = \int \Delta(t - t') f(t') dt'.$$

Now consider the time-discrete version

$$\frac{x_i - x_{i-1}}{h} - m x_{i-1} = \frac{\delta_{i,1}}{h}, \quad (9.19)$$

where we assume a discrete time axis $t = l h$ with $l \in [1, \dots, M]$. Assuming $x_{\leq 0} = 0$, derive an iterative expression for the solution and an explicit expression for $x_l = x(l h)$. Show that in the limit $h \rightarrow 0$ the result is identical to the analytical solution in continuous time. (2 points).

Now we want to express the Green's function with the help of a Gaussian integral. To this end write the iterative equation with the help of a Dirac- δ constraint. Show that we may define the generating function as

$$Z(j_1, \dots, j_M) := \prod_{l=1}^M \int_{-\infty}^{\infty} dx_l \delta(x_l - x_{l-1} - m x_{l-1} h - \delta_{l,1}) e^{j_l x_l h} \quad (9.20)$$

$$x_0 := 0$$

to obtain the Green's function $\Delta(k h)$ as $\frac{\partial Z}{\partial (h j_k)}$.

Show that Z is properly normalized

$$Z(0, \dots, 0) = 1. \quad (9.21)$$

We would like to bring Eq. (9.20) into the form of a Gaussian integral. We therefore introduce a second set of variables $\tilde{x}_l$ with $l \in [1, \dots, M]$ and show that the generating function with $\tilde{j} = (-\frac{1}{h}, 0, \dots, 0)$ can be brought to the form

$$Z(j_1, \dots, j_M, \tilde{j}_1, \dots, \tilde{j}_M) = \prod_{l=1}^M \left\{ \int_{-\infty}^{\infty} dx_l \frac{1}{2\pi i} \int_{-i\infty}^{i\infty} d\tilde{x}_l \right\} \quad (9.22)$$

$$\times \exp \left(\sum_{i=1}^M \tilde{x}_i (x_i - x_{i-1} - m x_{i-1} h) \right.$$

$$\left. + j_i x_i h + \tilde{j}_i \tilde{x}_i h \right).$$

Here, in addition, the source term $\tilde{j}_i \tilde{x}_i h$ was introduced.

Using the linearity in the x_i of the iteration (9.19), show that the Green's function is (Hint: Where has the initial condition gone in this notation?)

$$\begin{aligned} \Delta(k, h) &= \partial_{j_k} Z \left(j, \tilde{j}_1 = -\frac{1}{h}, \tilde{j}_{>1} = 0 \right) \Big|_{j=0} \\ &\stackrel{\text{linearity}}{=} - \frac{\partial^2 Z}{\partial(h j_k) \partial(h \tilde{j}_1)} \Big|_{j=\tilde{j}=0} \stackrel{\text{def. as}}{=} - \langle x_k \tilde{x}_1 \rangle. \end{aligned}$$

(2 points).

To expose the form of a Gaussian integral, now write equation (9.22) in symmetric form, introducing $y_i = \begin{pmatrix} x_i \\ \tilde{x}_i \end{pmatrix}$ and $\tilde{j}_i = \begin{pmatrix} j_i \\ \tilde{j}_i \end{pmatrix}$ and show that the result is a Gaussian integral of the form

$$\begin{aligned} Z(\tilde{j}) &= Z(j_1, \dots, j_M, \tilde{j}_1, \dots, \tilde{j}_M) \\ &= \prod_{l=1}^M \left\{ \int_{-\infty}^{\infty} dx_l \frac{1}{2\pi i} \int_{-i\infty}^{i\infty} d\tilde{x}_l \right\} \exp \left(-\frac{1}{2} y^T A y + \tilde{j}^T y \right), \\ A &= \begin{pmatrix} A_{xx} & A_{x\tilde{x}} \\ A_{\tilde{x}x} & A_{\tilde{x}\tilde{x}} \end{pmatrix}. \end{aligned}$$

Determine the entries of the sub-matrices A_{xx} , $A_{\tilde{x}x}$, $A_{x\tilde{x}}$, and $A_{\tilde{x}\tilde{x}}$ and show that the iterative solution $x = (x_1, \dots, x_M)$ solving (9.19) also obeys

$$A_{\tilde{x}x} x = (-1, 0, \dots, 0). \quad (9.23)$$

(2 points). It follows that the solution of the iteration effectively inverts the matrix $A_{\tilde{x}x}$.

Convince yourself that A is self-adjoint (symmetric). Show that the inverse of A is again of block-wise form

$$A^{-1} = \begin{pmatrix} 0 & A_{\tilde{x}x}^{-1} \\ A_{xx}^{-1} & 0 \end{pmatrix}$$

and that consequently by applying the result of (3.3) and using that we must have $Z(\tilde{j} = 0) = 1$ (9.21) we have

$$Z(j, \tilde{j}) = \exp(j^T A_{xx}^{-1} \tilde{j}). \quad (9.24)$$

(2 points). Argue with the linearity of the problem (9.23) that the iterative solution for (9.19) in the case

$$\frac{x_i - x_{i-1}}{h} - m x_{i-1} = -\frac{\tilde{j}_i}{h}$$

can be obtained from the above-determined solution as

$$x_i = \sum_{k=0}^i (1 + h m)^{i-k} \tilde{j}_k. \quad (9.25)$$

Furthermore, argue along the same lines with (9.23) that solution can also be seen as the inversion of

$$A_{\tilde{x}x} x = \tilde{j}. \quad (9.26)$$

With this insight, determine the explicit form of (9.24) and show that its limit $h \rightarrow 0$ and $M \rightarrow \infty$ is

$$Z[j, \tilde{j}] = \exp \left(\iint j(t) H(t - t') e^{m(t-t')} \tilde{j}(t') dt dt' \right). \quad (9.27)$$

So the Green's function can be obtained as

$$\Delta(t - t') = -\frac{\delta}{\delta j(t)} \frac{\delta}{\delta \tilde{j}(t')} Z \Big|_{j=\tilde{j}=0}.$$

(2 points).

(c) Propagators by Completion of Square

Perform the calculation from Eqs.(8.4) to (8.7) explicitly by completion of the square, following along the lines of the calculation in Sect.3.2 for the N -dimensional Gaussian. You may shift the integration variable as $y(t) \rightarrow y(t) + \int ds \Delta(t, s) \tilde{j}(s)$ and use that $Z(0) = 1$ is properly normalized. Also observe that Eq.(8.3) is self-adjoint, i.e. acting on the left argument in a scalar product has the same form (following from integration by parts and assuming vanishing boundary terms). (4 points).

(d) Unitarity of the Jacobian of the Time Evolution Operator

Prove that the Jacobian T' of the functional $T : x \rightarrow dx - f(x) dt$, where the right-hand side follows the Ito convention, has a unit determinant. To this end, consider discretization of the time axis, as in Chap. 7, and show that $\int \mathcal{D}x \delta[dx - f(x) dt] = 1$. Then use $\int \mathcal{D}x \delta[T[x]] \det(T') = 1$, which can be shown by the N -dimensional chain rule.

(e) Langevin Equation with Non-white Noise

In Chap. 8, we considered the case of a stochastic differential equation driven by white noise; its covariance function is $\propto \delta(\tau)$. In the coming lectures we will encounter effective equations of motion that are driven by a noise η that is non-white. An example is Eq. (10.19) that appears in the context of a random network. We want to see how such properties appear in the path-integral representation. To this end we extend this notion to the case of non-white noise, namely a stochastic differential equation

$$\partial_t x(t) = f(x(t)) + \eta(t), \quad (9.28)$$

where the driving noise is defined by its moment-generating functional

$$Z_\eta[j] = \left\langle e^{j^T \eta} \right\rangle_\eta = \exp \left(j^T m + \frac{1}{2} j^T c j \right),$$

where $j^T c j = \iint dt dt' j(t) c(t, t') j(t')$ and $c(t, t') = c(t', t)$ is symmetric. Determine all cumulants of η . What kind of process is it? (2 points).

By using the Fourier representation of the Dirac- δ -functional

$$\delta[x] = \int \mathcal{D}_{2\pi i} \tilde{x} \exp(\tilde{x}^T x),$$

derive the moment-generating functional $Z(j) = \langle e^{j^T x} \rangle$ for the process x that follows the stochastic differential equation (9.28). (2 points).

(f) Perturbative Corrections in a Non-linear SDE

We here consider the example of the non-linear stochastic differential equation from the lecture, similar to (9.3), but with a parameter $m < 0$ controlling the leak term

$$dx(t) = m x(t) dt + \frac{\epsilon}{2!} x^2(t) dt + dW(t), \quad (9.29)$$

where $\langle dW(t)dW(s) \rangle = D \delta_{t,s} dt$ is a Gaussian white noise process. First neglect the perturbation by the non-linear term by setting $\epsilon = 0$. What is the mean value $\langle x(t) \rangle$, the covariance $\langle\langle x(t)x(s) \rangle\rangle$, and the linear response of the mean value $\langle\langle x(t)\tilde{x}(s) \rangle\rangle$ of the process? Combine the latter two into the propagator matrix, once written in time domain, once in frequency domain. (2 points).

We would now like to obtain corrections to these cumulants by diagrammatic techniques. We already obtained the perturbation correction to first order in ϵ for the mean value in Sect. 9.2. Adapt the result for the SDE (9.29) (2 points). Determine the perturbative corrections to the covariance. Is there a correction to the covariance

at first order in ϵ ? Argue in terms of diagrams. (2 points). Then draw the diagrams that contribute to the covariance at lowest non-vanishing order in ϵ . (4 points). Write down the corresponding integrals in time domain (2 points) and in frequency domain (2 points).

Show that in frequency domain all diagrams decompose into the propagators attaching the two external lines $j(\omega)$ and a single loop, which amounts to a single frequency integral. Evaluate the integrals in frequency domain for all diagrams to obtain the correction to the power spectrum of the process, the propagator $\Delta_{xx}(\omega)$. (2 points).

References

1. C. De Dominicis, L. Peliti, Phys. Rev. B **18**, 353 (1978)
2. H.-K. Janssen, Zeitschrift für Physik B Condensed Matter **23**, 377 (1976)